Hard Level Questions

Programming & Open Source • 1st Year

Name: _____ Roll No: _____ Date: _____

Instructions: Answer all 5 questions. Show your reasoning wherever asked. The answer key is at the end of this document.

Section A: Questions

Q1. Emoji → Programming Language

Name the programming language each emoji represents.

				
(a)	(b)	(c)	(d)	(e)

Q2. Who Am I?

I started a hobby project in 1991 that now runs most of the world's servers and Android phones. I also created the version control tool everyone uses on GitHub.

Who am I, and what are the two creations?

Q3. Switch Puzzle

Three switches outside a closed room control three bulbs inside. You can enter the room only once. How do you find which switch controls which bulb?

Q4. The Binary Search That Never Ends (Python)

This code has **two bugs**.

```
def binary_search(arr, target):
    low, high = 0, len(arr) - 1
    while low < high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            low = mid
        else:
            high = mid
    return -1
```

Find both bugs, and give an input where the code hangs and another where it returns the wrong answer.

Q5. License Detective

Match each license to its key feature.

License	Key feature
MIT	A. Anyone who modifies it and offers it as a network service must share their source
GPL v3	B. Very short and permissive, only requires keeping the copyright notice
Apache 2.0	C. Derivative works that are distributed must also be GPL (copyleft)
AGPL	D. Permissive, but with an explicit patent grant

Section B: Answer Key

A1. Emoji → Programming Language

Emoji	Language	Why
	**Java**	The Java logo is a steaming cup of coffee
	**Ruby**	Ruby is a gemstone
	**Go**	Go's mascot is the Gopher
	**Swift**	Swift's logo is a swift bird
	**Perl**	Perl's mascot is a camel

A2. Who Am I?

Linus Torvalds. The two creations are **Linux** (the kernel behind most servers and Android) and **Git** (the version control tool GitHub is built around).

A3. Switch Puzzle

- Turn **switch 1** on and wait about 10 minutes.
- Turn **switch 1** off, then turn **switch 2** on.
- Enter the room (only once).

What you see	Switch
Bulb is **on**	Switch 2
Bulb is **off but warm**	Switch 1
Bulb is **off and cold**	Switch 3

A4. The Binary Search That Never Ends

- **Bug 1 (infinite loop):** `low = mid` and `high = mid` can leave the search range unchanged. They should be `low = mid + 1` and `high = mid - 1`.
- **Bug 2 (skipped case):** `while low < high` stops when one element is left, so that element is never checked. It should be `while low <= high`.

Input that hangs: `binary_search([1, 3, 5, 7], 7)`. Here low becomes 2 and high stays 3, so mid is always 2 and `low = mid` never moves forward.

Input with the wrong answer: `binary_search([5], 5)`. Here low = high = 0, so the loop never runs and the function returns -1 even though 5 is present.

Corrected code:

```
def binary_search(arr, target):
    low, high = 0, len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            low = mid + 1
        else:
            high = mid - 1
    return -1
```

A5. License Detective

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GPL v3	**C**	Copyleft: distributed derivative works must also be GPL
Apache 2.0	**D**	Permissive, with an explicit patent grant
AGPL	**A**	Network use counts as distribution; source must be shared